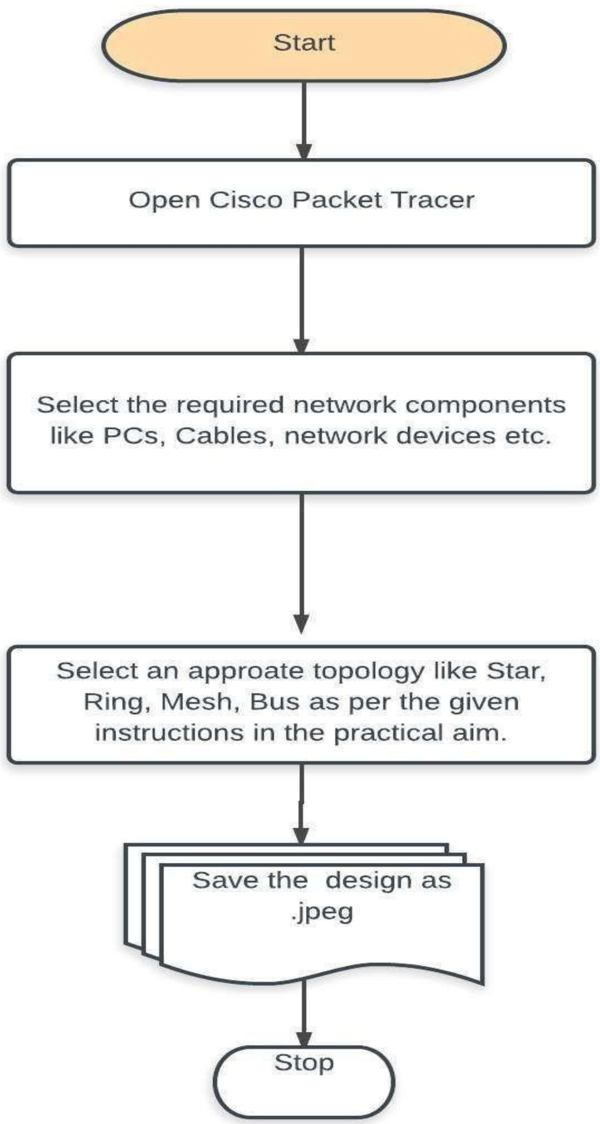
EXPERIMENT NO: 1

AIM: Experiments on Simulation Tools: (CISCO PACKET TRACER): To understand environment of CISCO PACKET TRACER to design simple network and perform experiments.

1.1 Purpose:

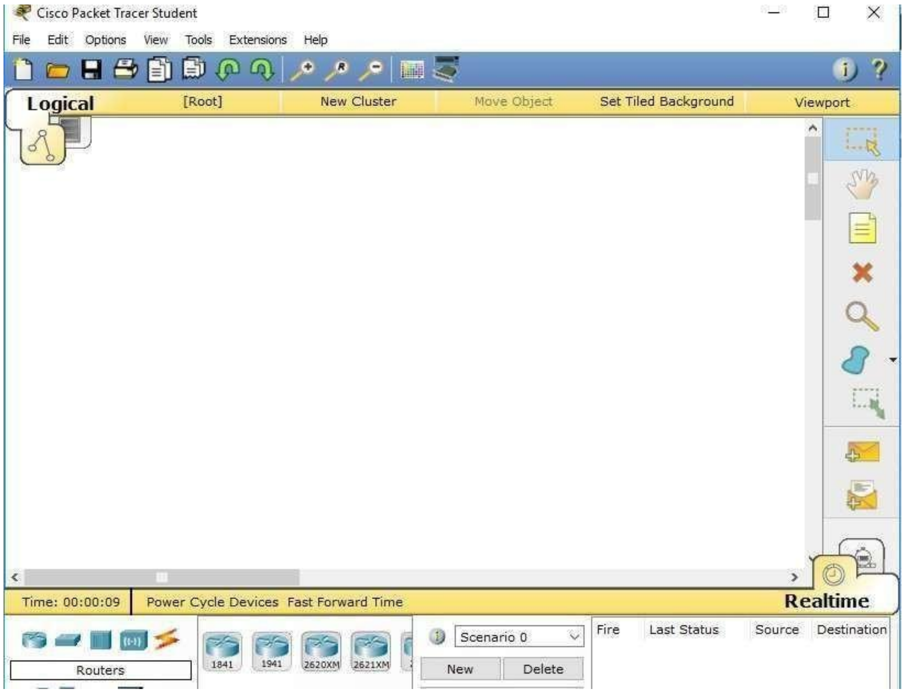
The main objective of the proposed experiment is to give exposure of numerous simulation devices to students such that student can actually design various network topologies in real-time.

1.2 Logical Flow of Practical

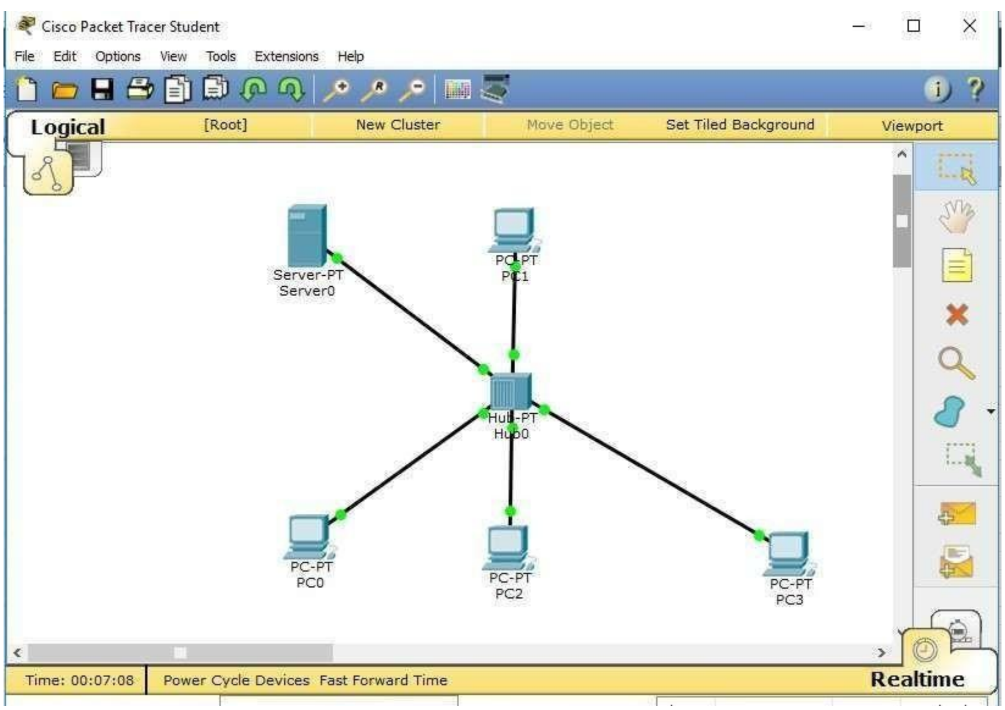


1.3 TOOLS/SOFTWARE:

Cisco Packet Tracer

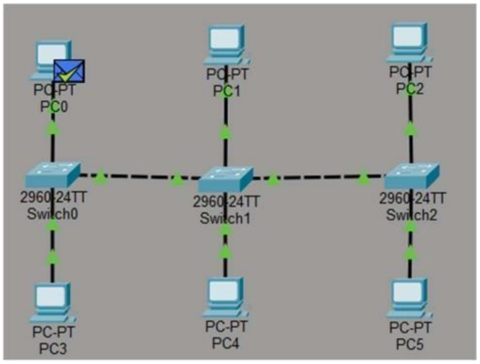


1.4 Expected Output:

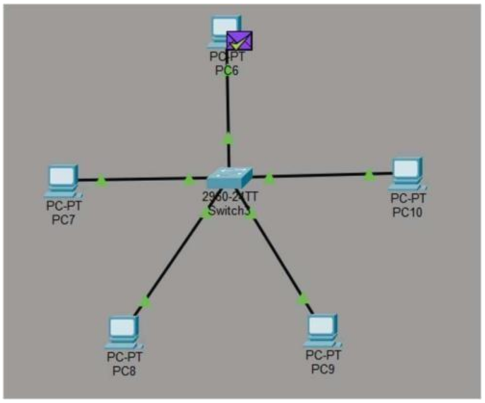


1.5 Practice Exercises:

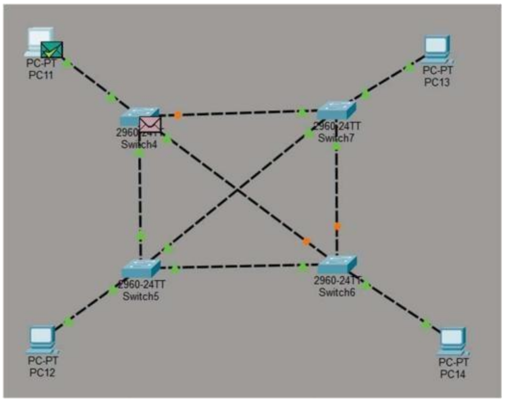
1. **Design a Bus topology network using Cisco Packet Tracer resources.**



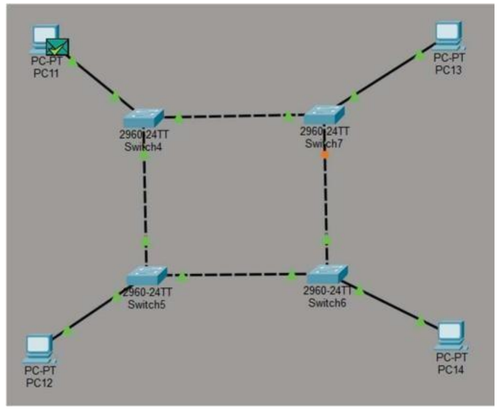
2. **Design a Star Topology network using Cisco Packet Tracer resources.**



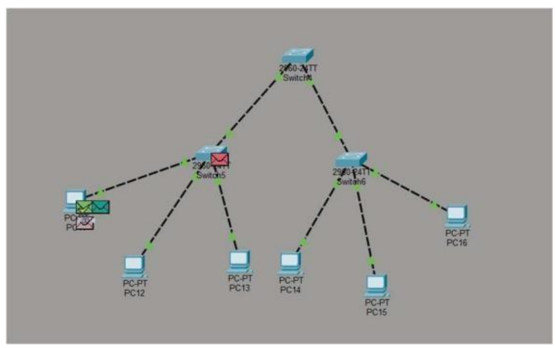
3. **Design a Mesh Topology network using Cisco Packet Tracer resources.**



4. **Design a Ring Topology network using Cisco Packet Tracer resources.**



5. **Design a Tree Topology network using Cisco Packet Tracer resources.**



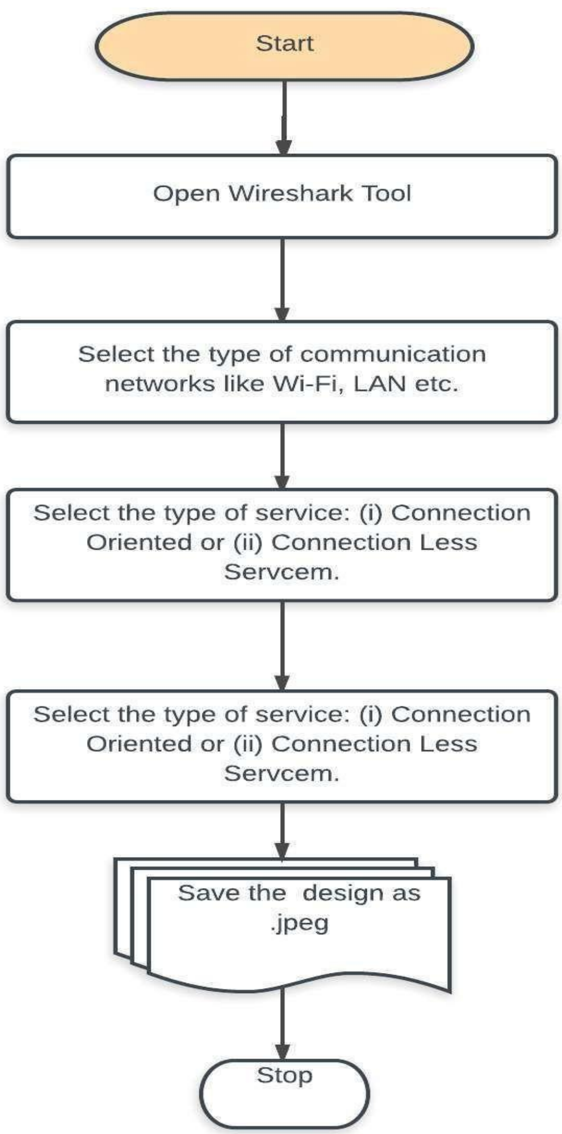
EXPERIMENT NO: 2

AIM: To understand the features of wireshark as a packet capture tool and understand encapsulation of information. Also study effect of few network commands

2.1 Purpose:

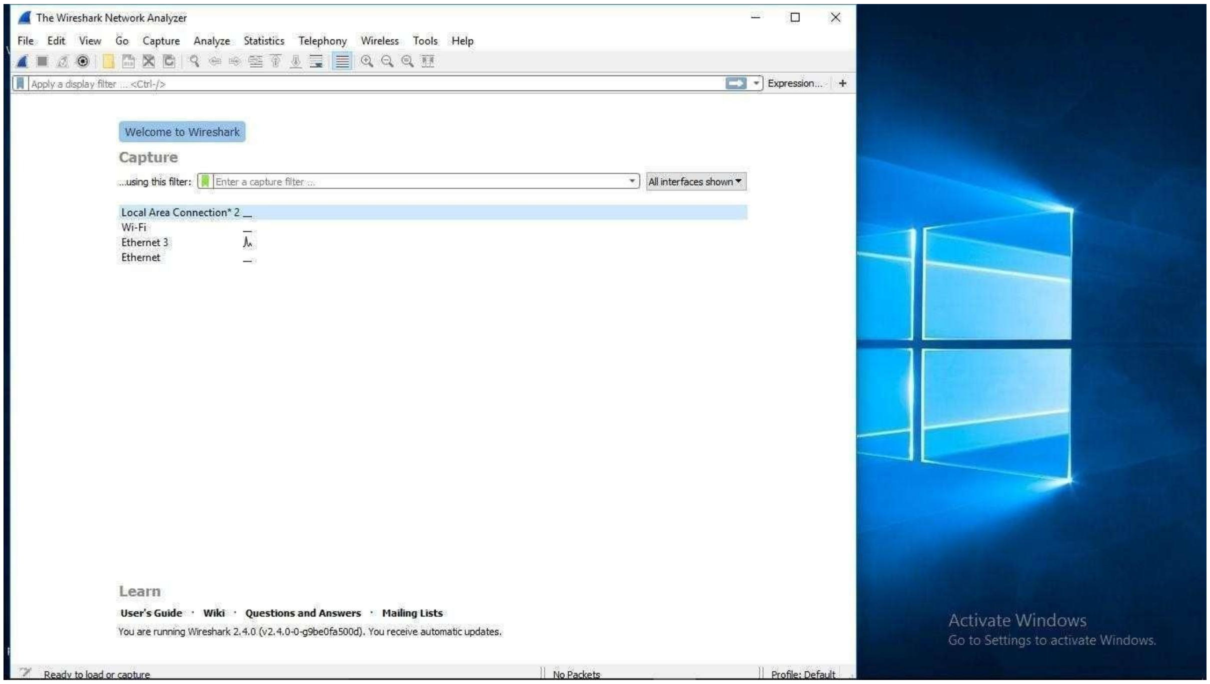
The main objective of the proposed experiment is to give exposure of wire-shark tool to students so students can learn to monitor transmission packets being sent in Wi-Fi and LAN environments.

2.2 Logical Flow of Practical

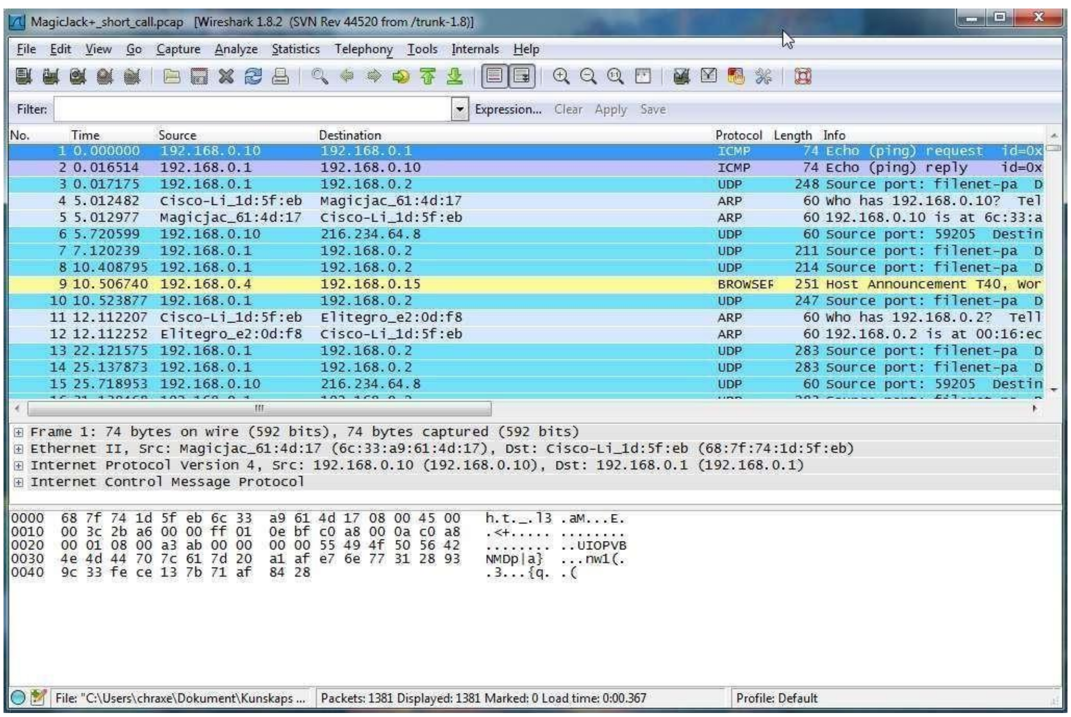


2.3 TOOLS/SOFTWARE:

Wireshark



2.4 Expected Output:



2.5 Practice Excercises:

1. Monitor TCP packets in a LAN connection and record your observations.
2. Monitor UDP packets in a LAN connection and record your observations.
3. Monitor TCP packets in a Wi-Fi connection and record your observations.
4. Monitor UDP packets in a Wi-Fi connection and record your observations.

EXPERIMENT NO: 3

AIM: To study behaviour of generic devices used for networking:

Design a simple network with multiple nodes and connect via generic devices available in library. Perform simulation and trace communication behaviour of specified network devices.

1: Use only HUB to design a small network having 4 to 6 hosts

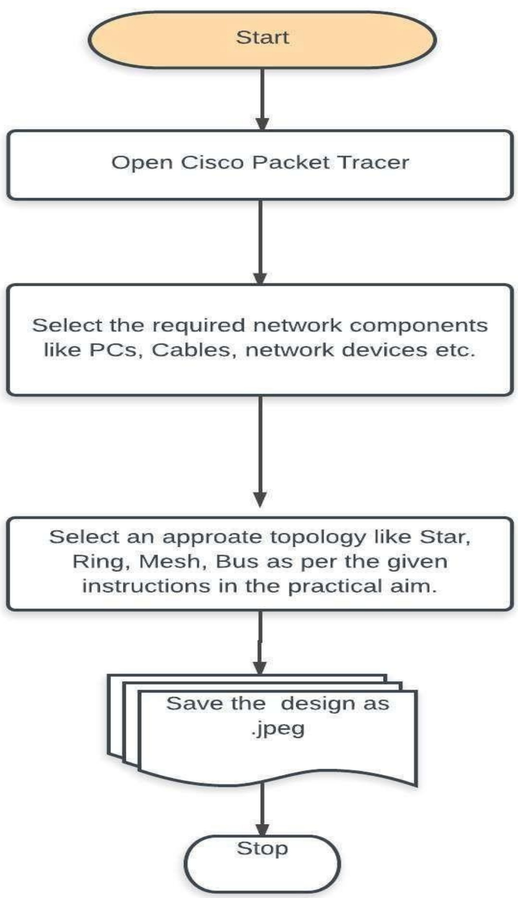
2: Use only a Switch to design a small network with 4 to 6 hosts.

3: Use both the device (HUB and SWITCH) for a network and find out functioning difference between switch and hub.

1.1 Purpose:

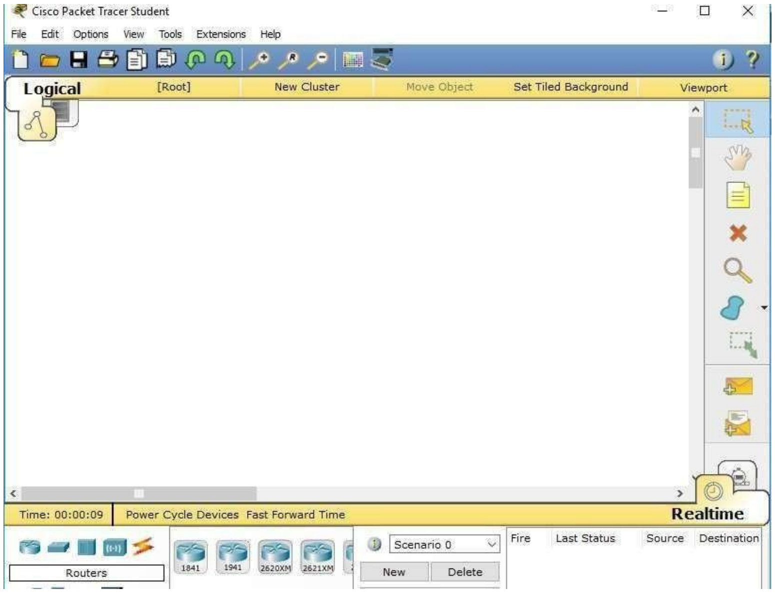
The main objective of the proposed experiment is to give exposure of numerous simulation devices to students such that student can actually design various network topologies in real-time using Hub and Switch network devices.

1.2 Logical Flow of Practical

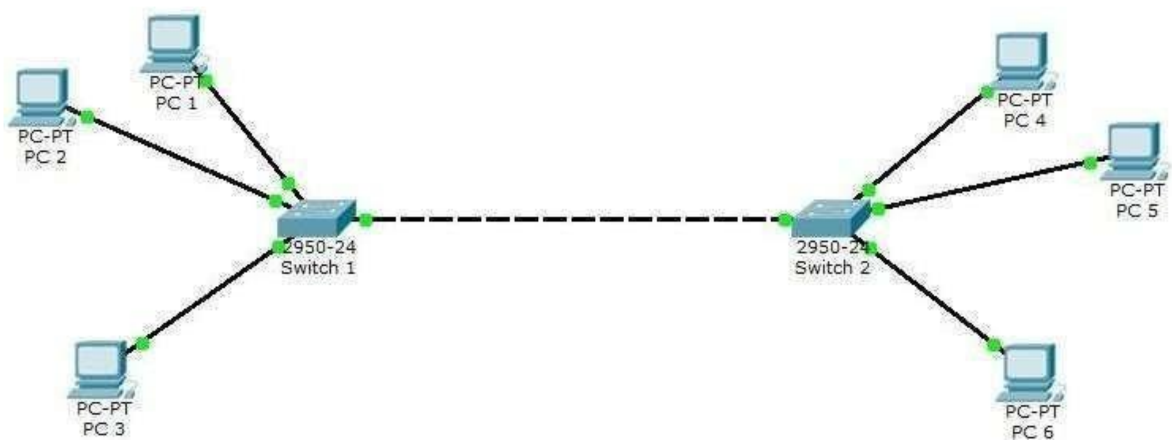


1.3 TOOLS/SOFTWARE:

Cisco Packet Tracer



1.4 Expected Output:



1.5 Practice Excercises:

1. Design a Star and Bus topology network. Connect the designed network using multiple switches.

3. Design a Bus and Mesh Topology network using Cisco Packet Tracer resources.

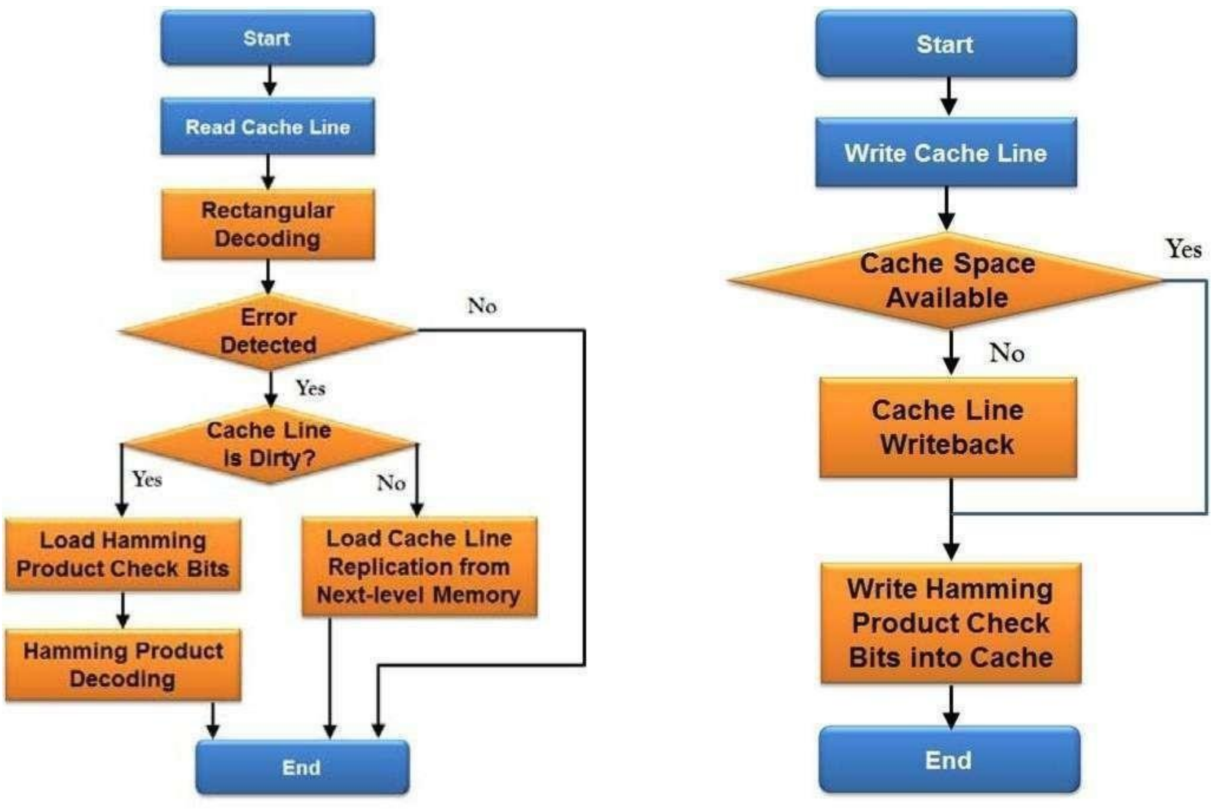
EXPERIMENT NO: 4

AIM: Data Link Layer (Error Correction): Write a program to implement error detection and correction using HAMMING code concept. Make a test rum to input data stream and verify error correction feature.

4.1 Purpose:

The main objective of the proposed experiment is to understand error correction between a sender and a receiver using a hamming code concept.

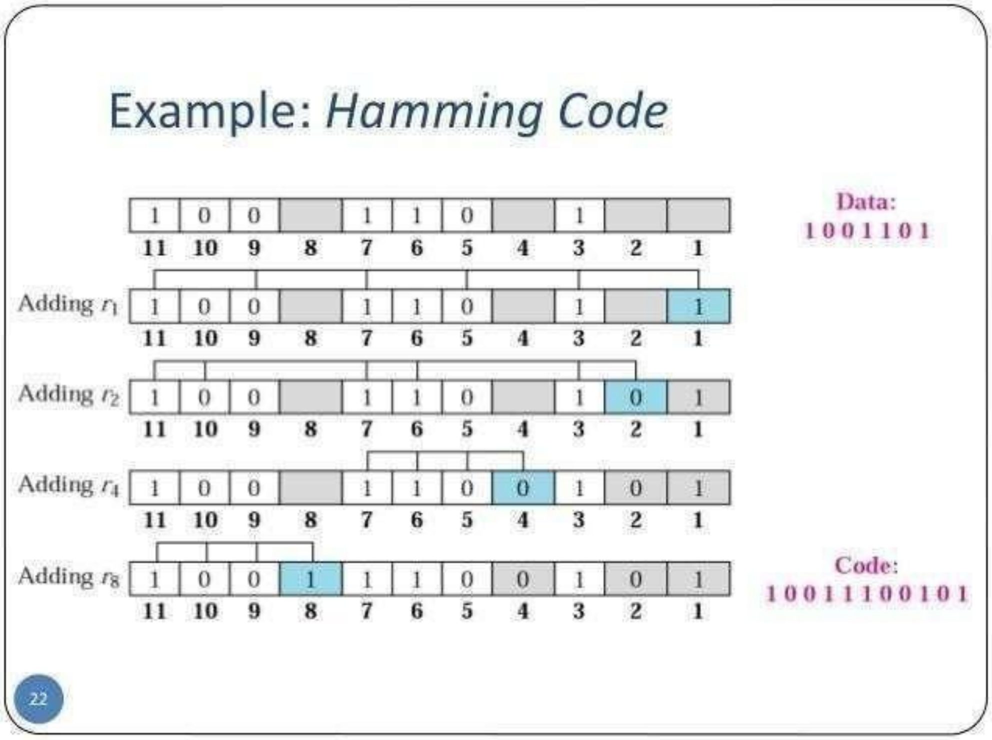
4.2 Logical Flow of Practical



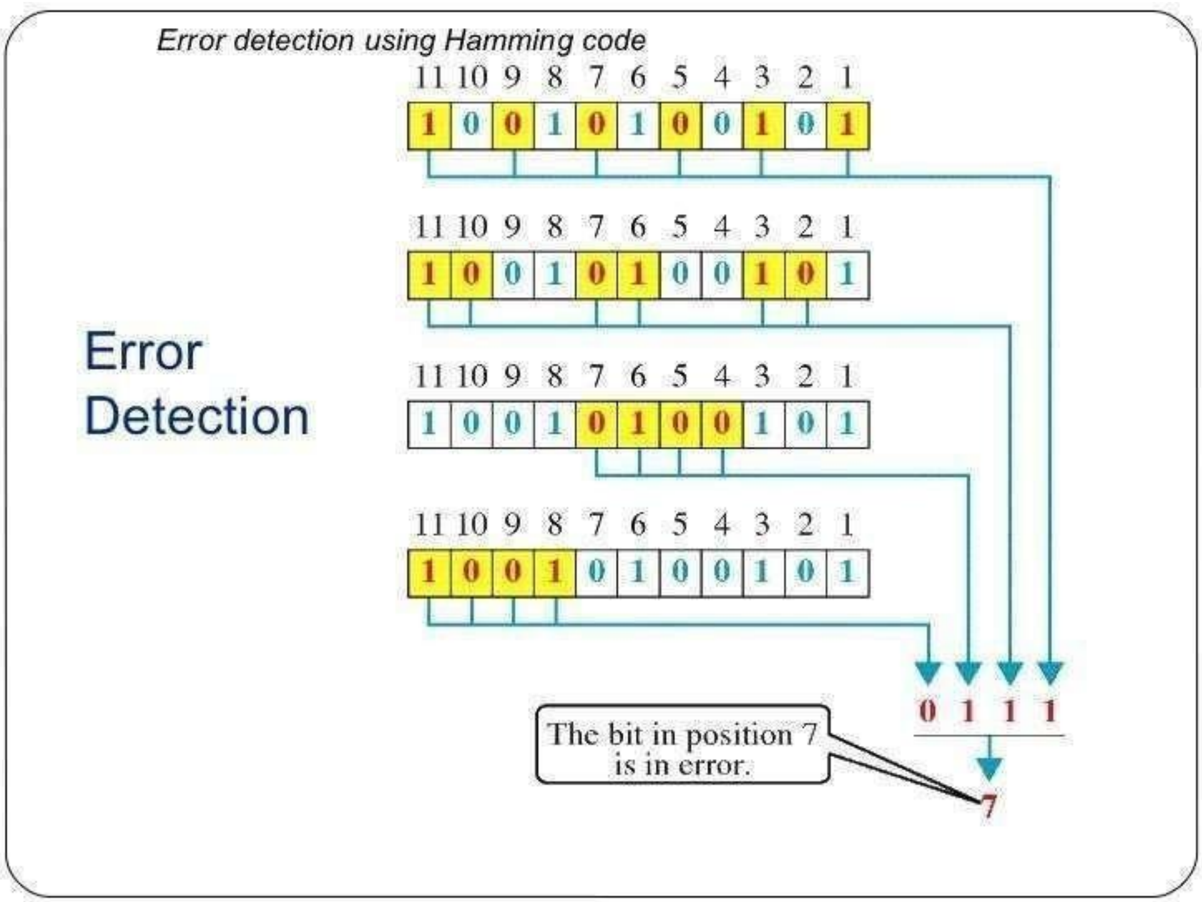
(a) (b)

4.3 TOOLS/SOFTWARE:

Language C



4.4 Expected Output:



4.5 Practice Excercises:

1. Apply hamming code error correction technique in any packet switching network.
2. Apply hamming code error correction technique in any circuit switching network.

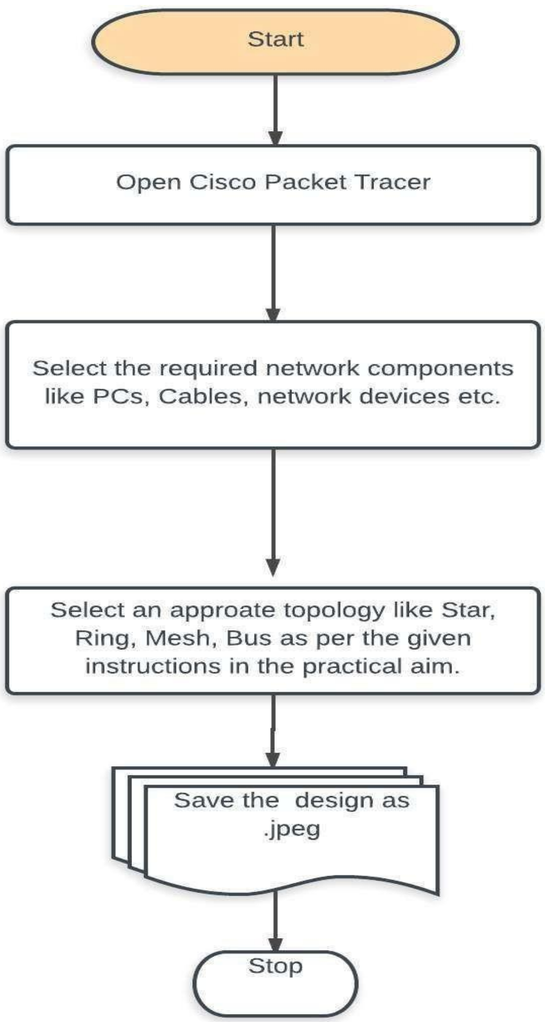
EXPERIMENT NO: 5

AIM: Virtual LAN: Simulate Virtual LAN configuration using CISCO Packet Tracer Simulation.

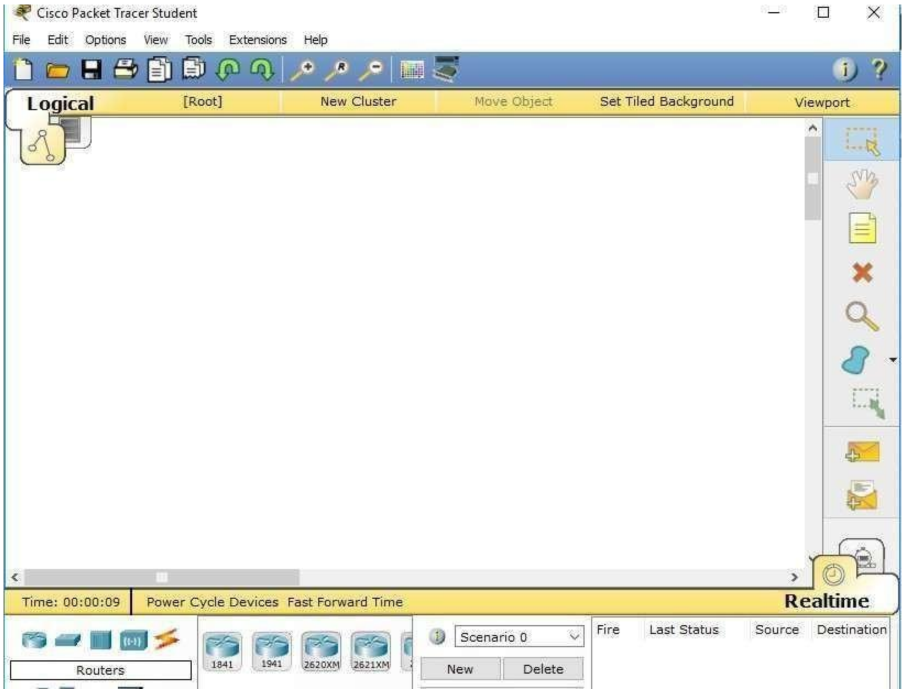
5.1 Purpose:

The main objective of the proposed experiment is to establish a virtual private network using Cisco Packet Tracer resources and learn various LAN settings. Also design various wireless network topologies in real-time using Hub and Switch network devices.

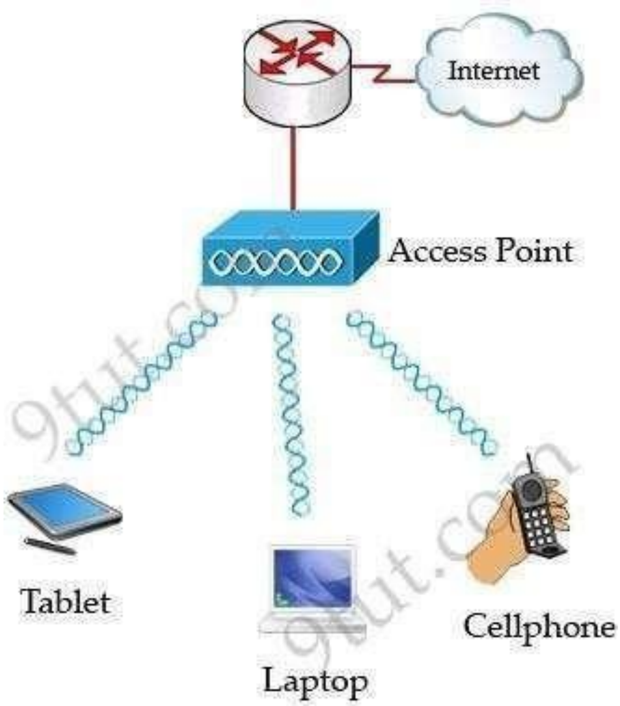
5.2 Logical Flow of Practical



5.3 TOOLS/SOFTWARE:
Cisco Packet Tracer



5.4 Expected Output:



5.5 Practice Excercises:

1. Design a Wireless Star and Bus topology network. Connect the designed network using multiple switches.
2. Design a Wireless Bus and Mesh Topology network using an access point.

EXPERIMENT NO: 6

AIM: Wireless LAN: Configuration of Wireless LAN using CISCO Packet Tracer

6.1 Purpose:

The main objective of the proposed experiment is to establish wireless LAN network using Cisco Packet Tracer resources, learn various LAN settings and simulation network using Router and Switch network devices.

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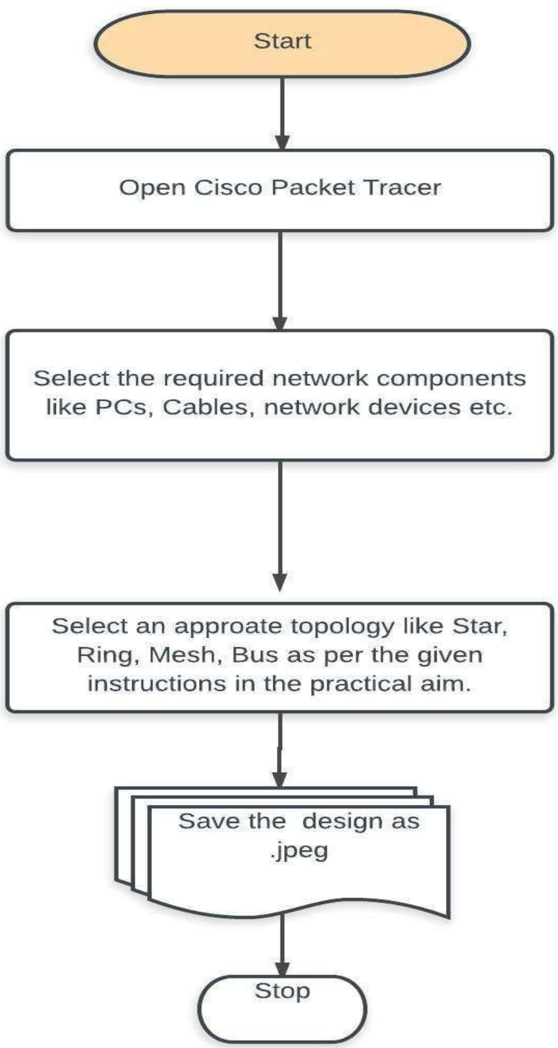
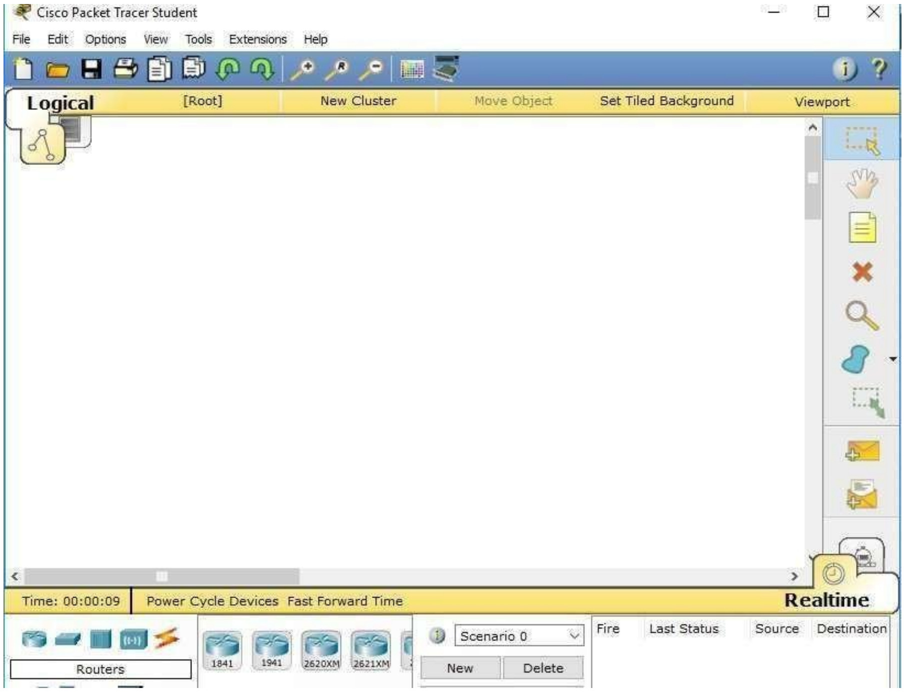
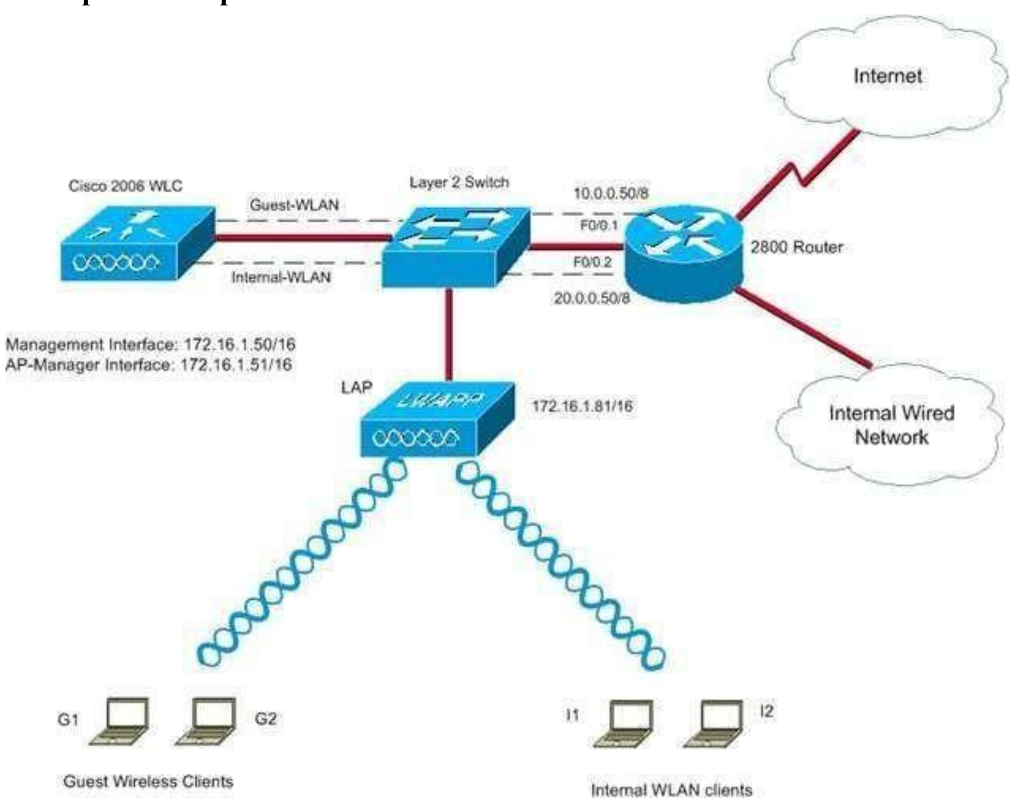
6.2 Logical Flow of Practical



6.3 TOOLS/SOFTWARE:
Cisco Packet Tracer



6.4 Expected Output:



6.5 Practice Excercises:

1. Design a Wireless LAN using a wireless router, nodes and access point.
2. Design a VPN network using an access point and a wireless router.

EXPERIMENT NO: 7

AIM: Internetworking with routers: Design a three or four simple networks (with 3 to 4 hosts) and connect via Router. Perform simulation and trace how routing is done in packet transmission.

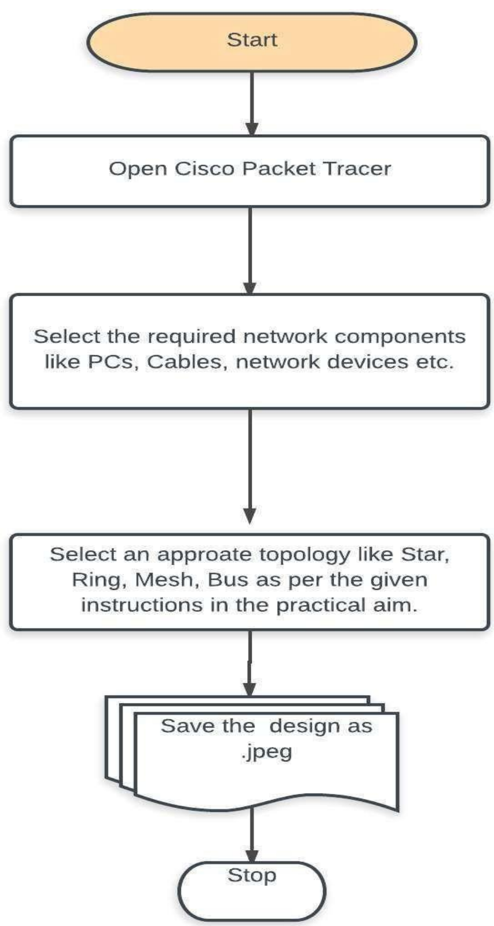
1: Experiment on same subnet

2: Perform Experiment across the subnet and observe functioning of Router via selecting suitable pair of Source and destination.

7.1 Purpose:

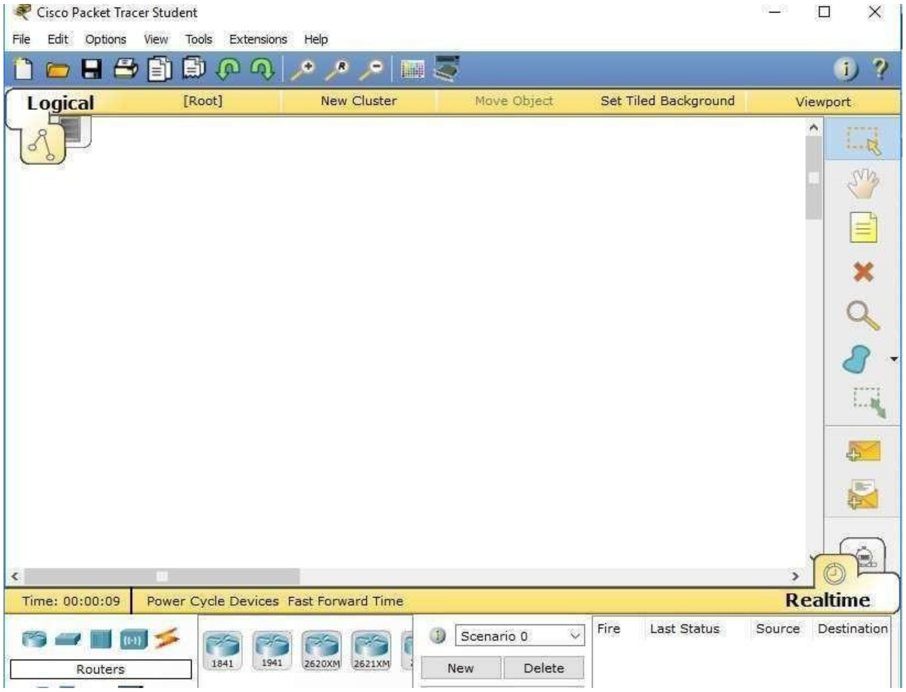
The main objective of the proposed experiment is to connect multiple networks using a router. It also gives an opportunity to students to learn router configuration in multiple network environments.

7.2 Logical Flow of Practical

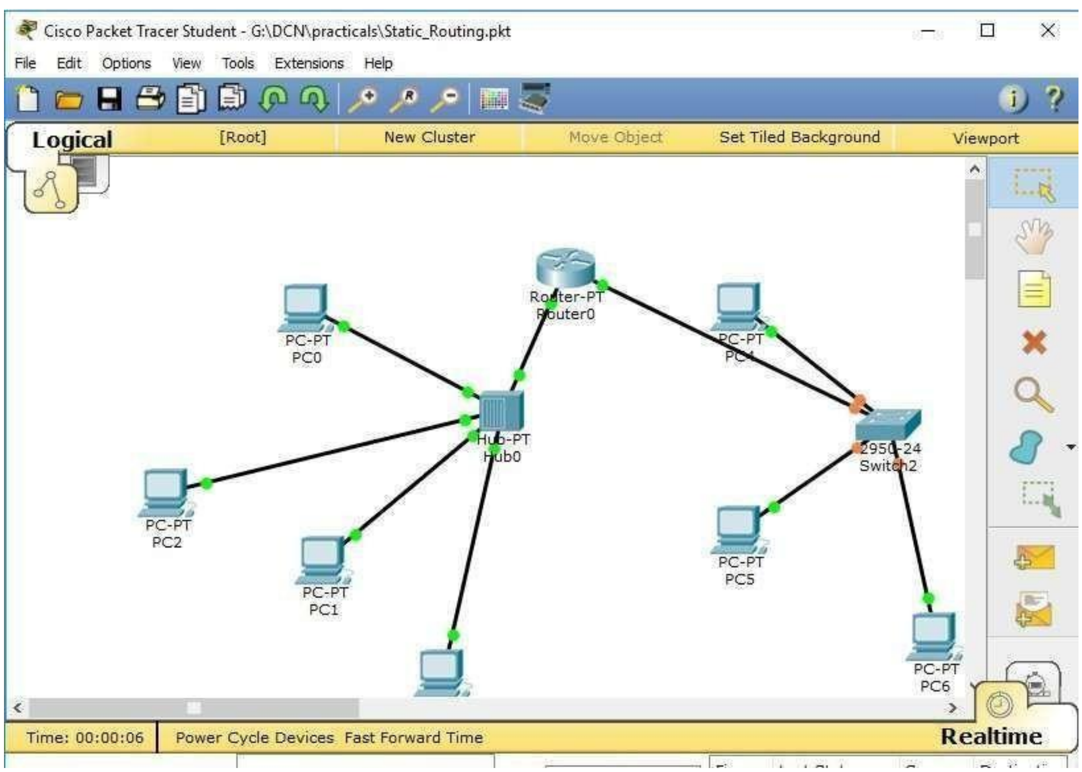


7.3 TOOLS/SOFTWARE:

Cisco Packet Tracer



7.4 Expected Output:



7.5 Practice Excercises:

1. Design and connect multiple networks using a serial router connection.

2. Design and connect multiple networks using a parallel router connection

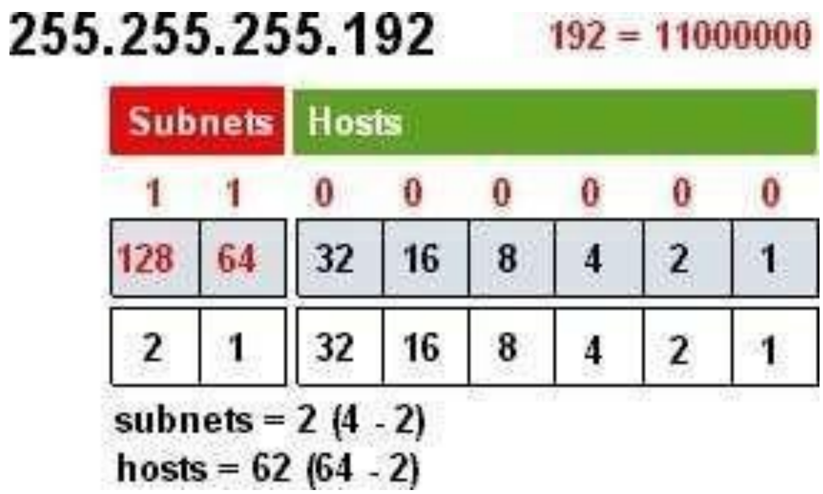
EXPERIMENT NO: 8

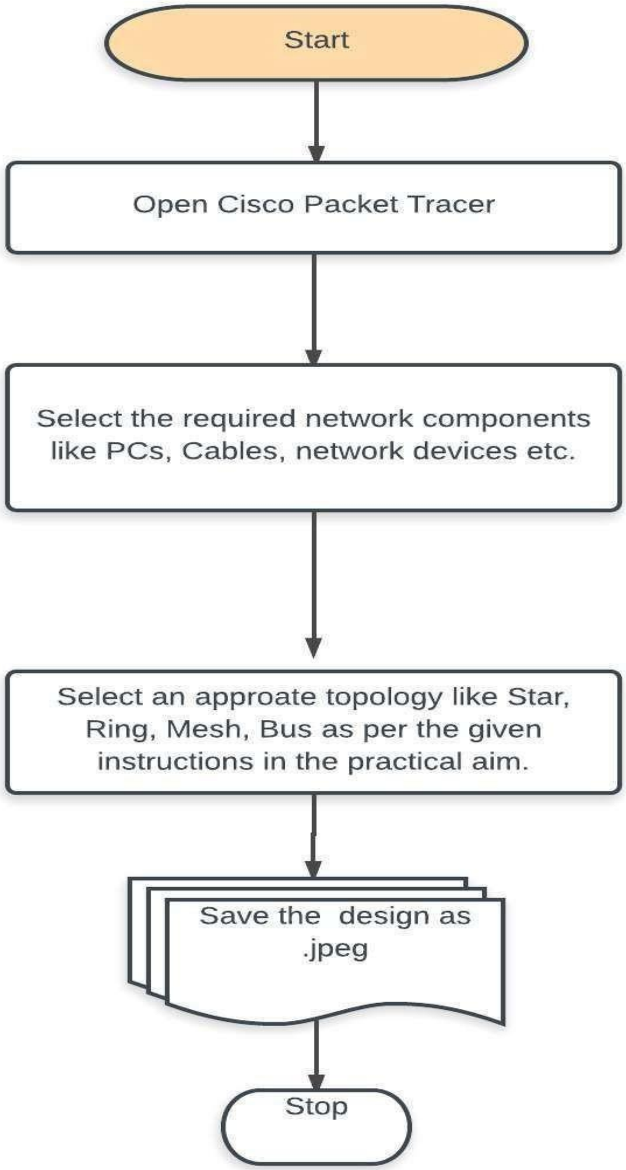
AIM: Implementation of SUBNETTING: Design multiple subnet with suitable number of hosts. Make a plan to assign static IP addressing across all subnet to explain implementation of SUBNETTING.

8.1 Purpose:

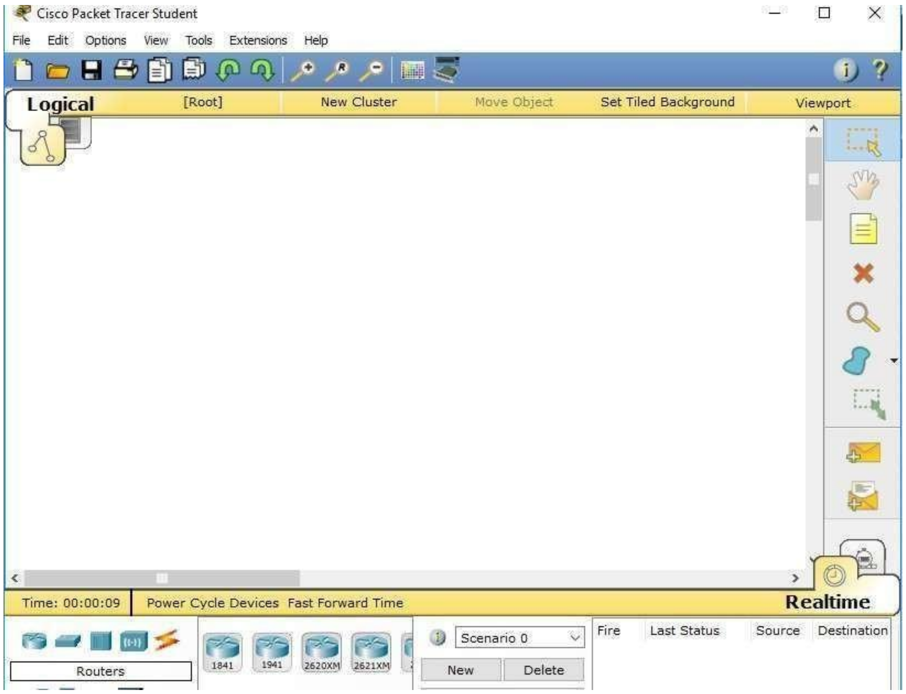
The main objective of the proposed experiment is to divide multiple networks using the concept of subnetting and super-netting.

8.2 Logical Flow of Practical

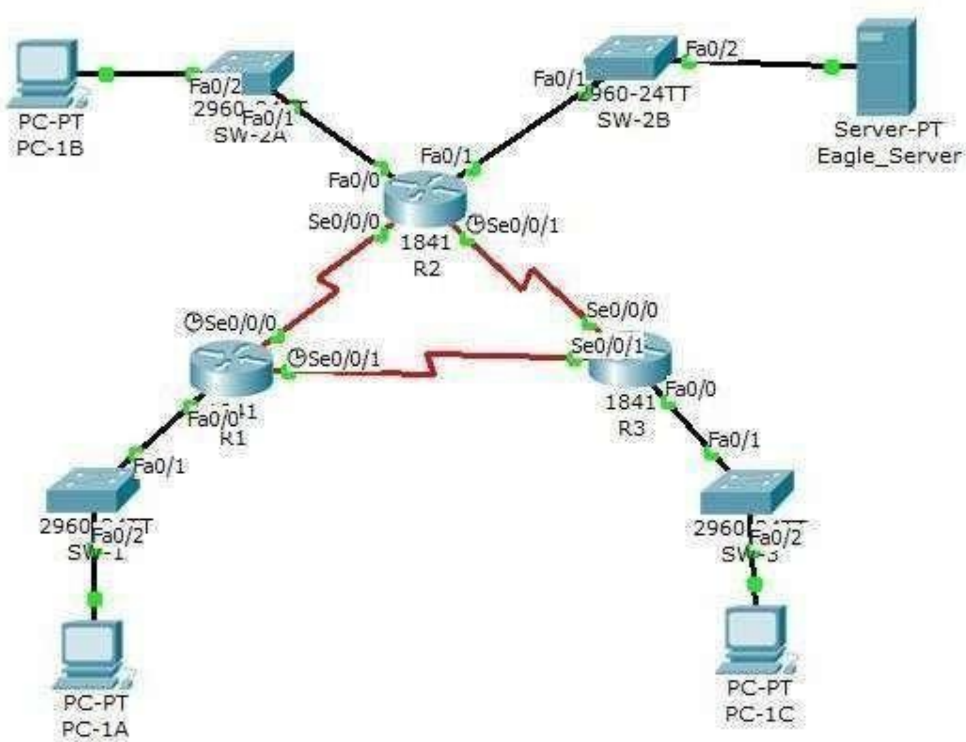




8.3 TOOLS/SOFTWARE:
Cisco Packet Tracer



8.4 Expected Output:



8.5 Practice Excercises:

1. Design and connect multiple networks using a serial router connection.

2. Design and connect multiple networks using a parallel router connection

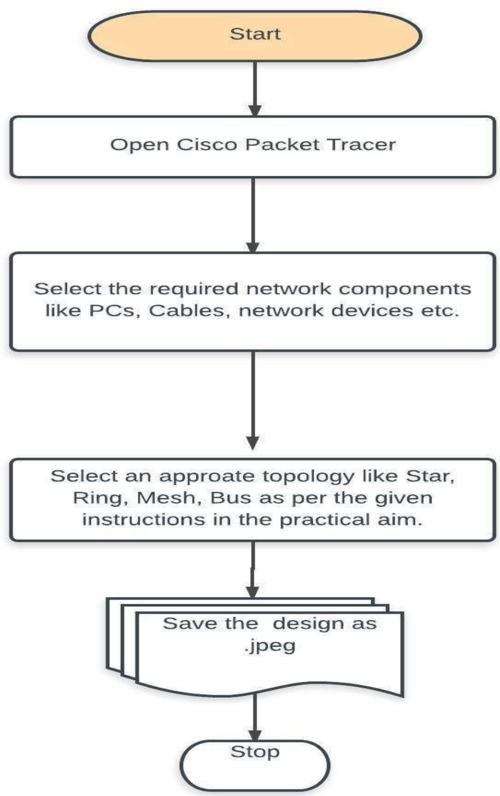
EXPERIMENT NO: 9

AIM: Routing at Network Layer: Simulate Static and Dynamic Routing Protocol Configuration using CISCO Packet Tracer.

9.1 Purpose:

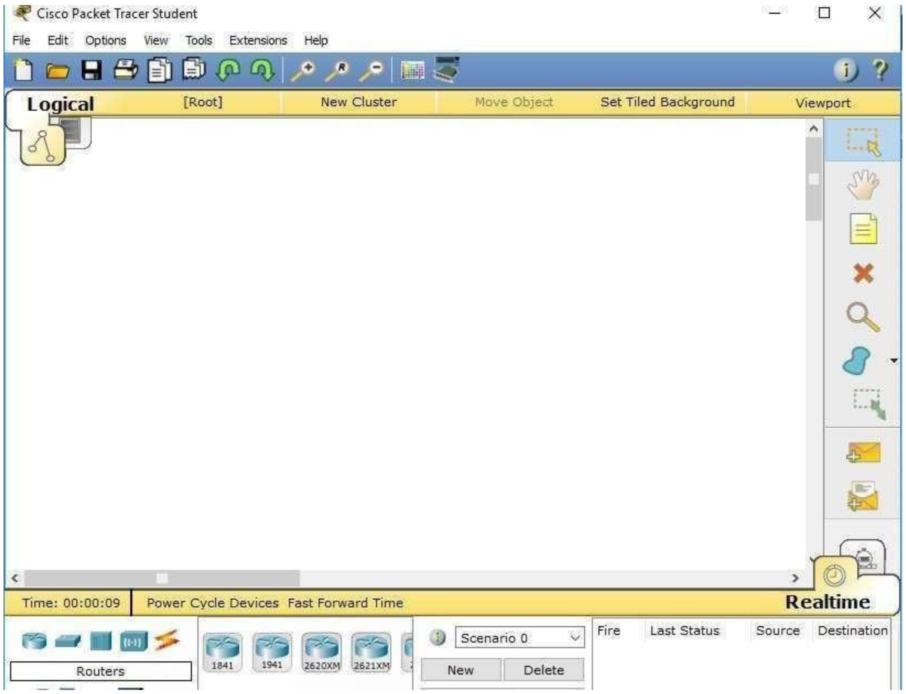
The main objective of the proposed experiment is to implement static and dynamic routing protocol using serial and parallel routing connections. It also gives an opportunity to students to learn serial and parallel router configuration in multiple network environments.

9.2 Logical Flow of Practical

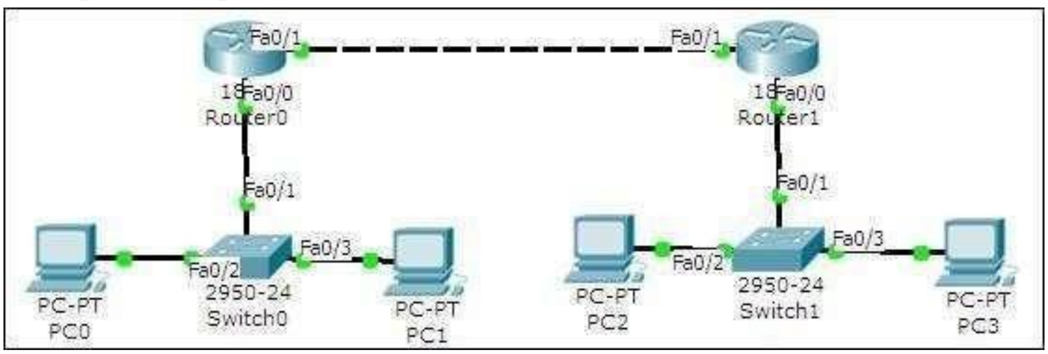


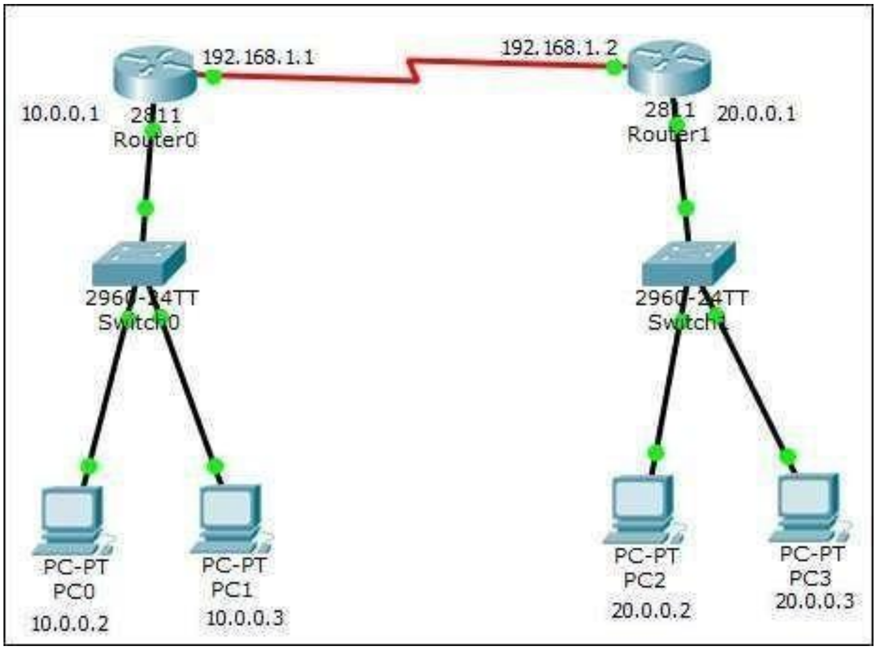
9.3 TOOLS/SOFTWARE:

Cisco Packet Tracer



9.4 Expected Output:





9.5 Practice Excercises:

1. Design an actual LAN network using 4 nodes. Design two subnets from the designed network using class A Ip address classes.

2. Design actual Wi-Fi network using 4 nodes. Design two subnets from the designed network using class A Ip address classes.

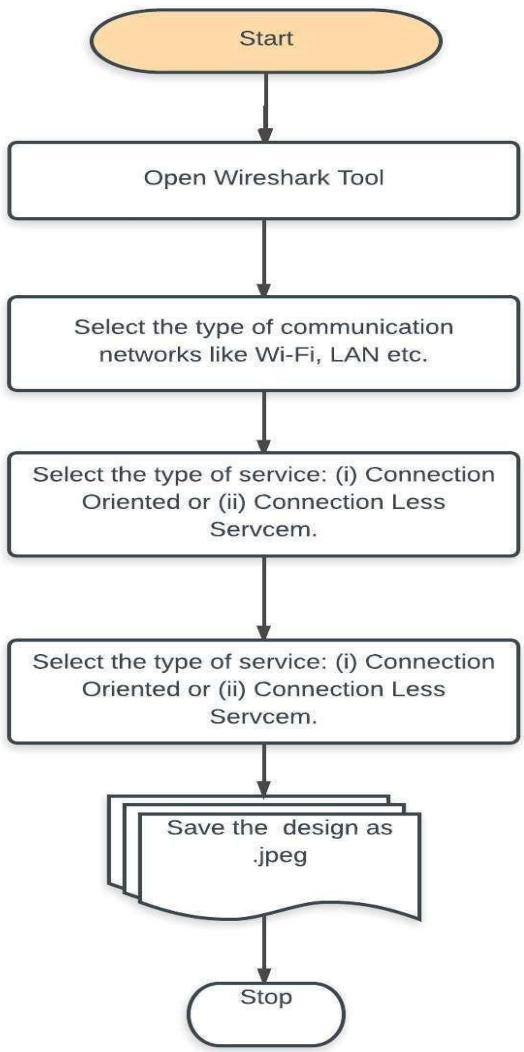
EXPERIMENT NO: 10

AIM: Experiment on Transport Layer: Implement echo client server using TCP/UDP sockets.

10.1 Purpose:

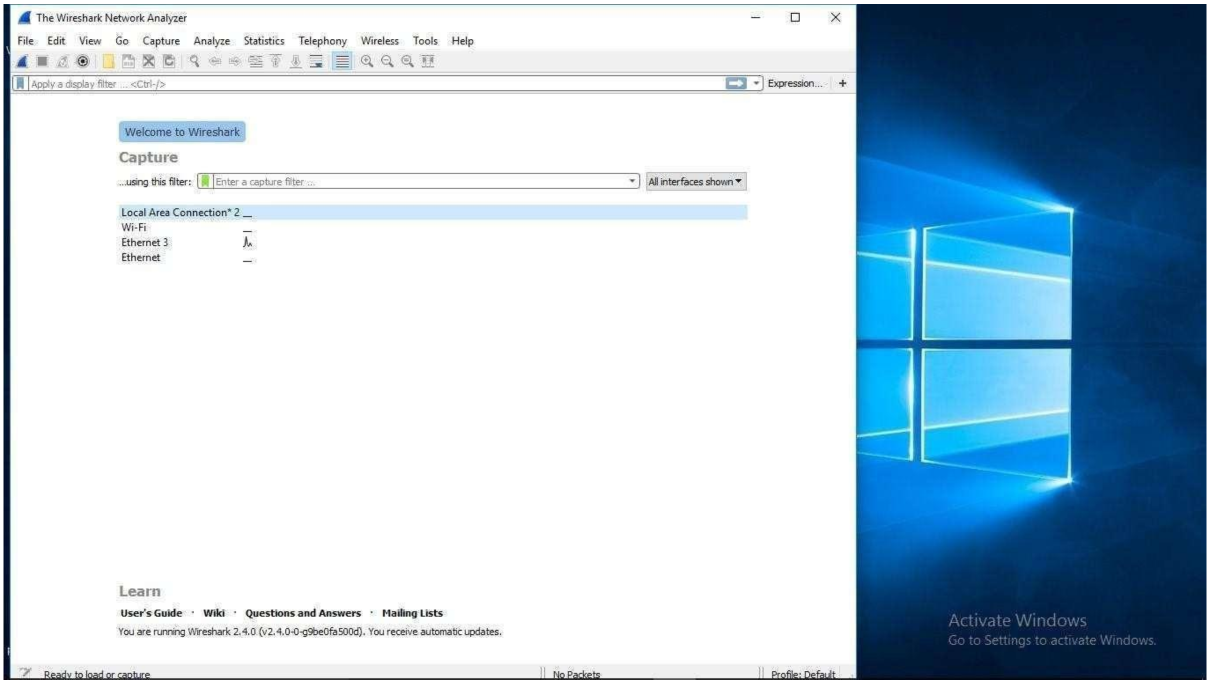
The main objective of the proposed experiment is to initiate communication between client and server nodes using echo client and echo server. Utilize both TCP and UDP sockets in the proposed experiment.

10.2 Logical Flow of Practical

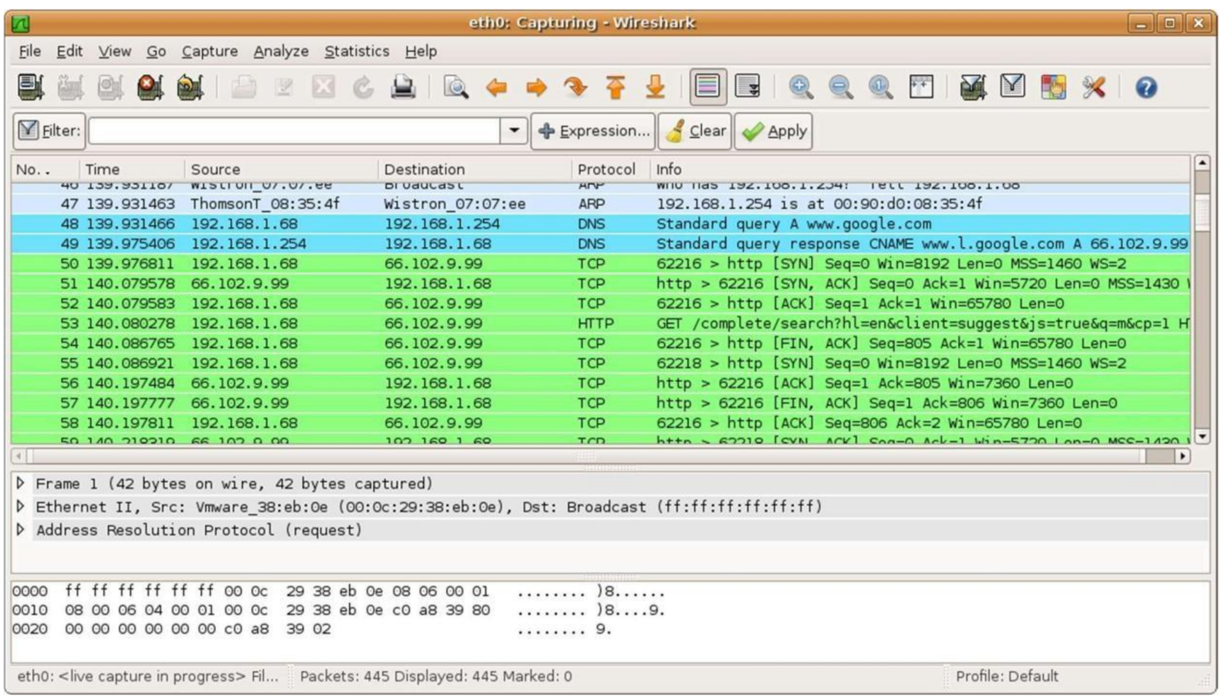


10.3 TOOLS/SOFTWARE:

Wireshark



10.4 Expected Output:



10.5 Practice Excercises:

1. **Test asynchronous and synchronous communication using wireshark in LAN environment.**
2. **Test asynchronous and synchronous communication using wireshark in Wi-Fi environment.**